Date: Day:

COAL Assignment 03

Samiullah Ashfaq SP24-BSE-107

(i)

MIPS :

MIPS is a measure of computer's processor speed that indicates how many millions of instructions the CPU can execute in one second. It is calculated by dividing the total instruction count by the execution time and then normalizing it to millions. However, MIPS is often criticised because it doesn't account for the complexity of different instruction sets across architectures.

$$\text{MIPS} = \frac{\text{Instruction Count}}{\text{Execution Time} \times 10^6}$$

$$\text{MIPS} = \frac{300,000,000}{2 \times 1,000,000}$$

$$= \frac{300}{2} = 150 \text{ MIPS}$$

Date:_____ Day:_____

(ii)

SPEC Benchmark:

SPEC (Standard Performance Evaluation Corporation) is a non-profit organization that provides standardized sets of benchmarks to evaluate the performance of the newest generation of computing systems. It measures performance by running a suite of real-world applications (like compilers, weather modeling or video compression) and comparing the execution time against a refrence machine.

It is more reliable than MIPS or MFLOPS because those metrics only measure raw instruction throughput, which doesn't necessarily translate to real-world speed. SPEC accounts for the entire system's efficiency, including the compiler, memory hierarchy, and I/O. By using a geometric

Date: Day:

mean of multiple tests, it prevents a single high-performing area from skewing the results, providing a holistic view of actual CPU capability.

MFLOPS :

MFLOPS is a metric specifically used to measure the performance of a processor in executing floating-point operations (math involving decimals). Unlike MIPS, which counts all instructions, MFLOPS focuses on the scientific and mathematical workload, making it a vital metric for supercomputing, graphics rendering and Ai applications.

$$\text{MFLOPS} = \frac{\text{Floating Point Operations}}{\text{Execution Time} \times 10^6}$$

$$\text{MFLOPS} = \frac{2,000,000,000}{4 \times 1,000,000} = \frac{2000}{4}$$

$$= 500\text{MFLOPS}$$